AyuAware Shop Section

Technical Product Requirements Document

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1. Executive Summary

What We're Building

The Shop section is the **Ayurvedic Marketplace** — a curated shopping experience where users find supplements, foods, herbs, and wellness products recommended specifically for their constitution, current imbalances, active protocols, and seasonal needs. It answers the fundamental question: *"What products will actually help ME based on my unique Ayurvedic profile?"*

This isn't generic e-commerce with wellness products. It's a personalized shopping experience that:

Dimension	What We Offer	Why It Matters
Twin-Guided Shopping	Digital Twin explains WHY each product is recommended	Users understand the purpose, not just the purchase
Ayurvedic Context	Every product shows traditional properties (Rasa, Virya, Vipaka)	Products understood through Ayurvedic lens
Protocol	Products from active Projects/Protocols	Shopping supports active health
Seasonal	Recommendations adjust with Ritucharya	Right products for right season
Constitutional Matching	Dosing and selection based on Prakriti/Vikriti	Personalized to individual constitution
Quality	Manufacturing standards, testing,	Trust through transparency

Why It Matters

Most supplement and wellness shops are generic. They list products by category with no personalization beyond "bestsellers" and "trending." Users have no idea if a product is appropriate for their constitution, will interact with their current supplements, or is even relevant to their health goals.

Ayurveda has sophisticated understanding of how substances affect different body types. The same herb can be medicine for one person and aggravating for another. Ashwagandha, for example, is excellent for Vata imbalance but can increase Pitta if taken inappropriately.

Key Differentiator

Generic supplement shop: "Ashwagandha - $24.99. Bestseller. Good for stress. Add to cart."

AyuAware Shop: "Ashwagandha — EXCELLENT MATCH for you. Your elevated Vata (40%) and active Nidra Protocol mean this herb directly addresses your needs. It calms Vata, supports healthy cortisol, and promotes restful sleep. For your Pitta-Vata constitution, take 1 capsule in the evening with warm milk. Note: It's warming, so monitor for excess heat. This pairs well with Brahmi, which you don't have in your stack yet."

Personalized. Ayurvedically contextualized. Protocol-integrated. Twin-guided.

2. Section Overview

2.1 Information Architecture

```
Shop Section
│
├── 🛒 Main Shop View
│    ├── Twin Guidance Header
│    ├── Search & Filters
│    ├── Tab Navigation (For You, Protocols, Seasonal, Browse
All)
│    ├── What Your Body Needs Today
│    ├── From Your Protocols
│    ├── Seasonal Picks
│    ├── Shop by Ayurvedic Action
│    └── Browse Categories
│
├── 📦 Product Categories
│    ├── Supplements & Herbs
│    │    ├── By Ayurvedic Action
│    │    ├── By Health Goal
│    │    └── By Dosha
│    ├── Foods & Ingredients
│    │    ├── Dosha-Balancing Foods
│    │    ├── Seasonal Foods
```

```
│       │       └── Superfoods
│       ├── Oils & Body Care
│       ├── Teas & Beverages
│       ├── Kitchen & Tools
│       └── Books & Resources
│
├── 📄 Product Detail View
│       ├── Product Image Gallery
│       ├── "Is This Good For You?" Assessment
│       ├── Ayurvedic Properties
│       ├── Dosage & Timing (Personalized)
│       ├── Product Details
│       ├── Quality Indicators
│       ├── Related Products
│       └── Chat Input
│
├── 🧺 My Stack (Cart)
│       ├── Current Stack Items
│       ├── Twin Suggestions
│       ├── Supplement Schedule Preview
│       └── Checkout
│
├── ♡ Wishlist
│       └── Saved Products
│
└── 📋 Orders & History
        ├── Active Orders
        ├── Order History
        └── Reorder
```

2.2 Product Categories

Supplements & Herbs:

Subcategory	Examples	Count
By Ayurvedic Action	Rasayana, Deepana, Medhya, Nidra, Balya, Ojas	~150
By Health Goal	Sleep, Digestion, Energy, Immunity, Stress	~100
By Dosha	Vata Balancing, Pitta Cooling, Kapha Reducing	~80

Foods & Ingredients:

Subcategory	Examples	Count

Dosha-Balancing	Foods suited to each constitution	~80
Seasonal Foods	Current season recommendations	~40
Superfoods	Ghee, honey, spices, grains	~40

Oils & Body Care:

Subcategory	Examples	Count
Massage Oils	Sesame, coconut, herbal oils	~30
Skincare	Face oils, body care	~30
Nasya Oils	Nasal administration oils	~15

Teas & Beverages:

Subcategory	Examples	Count
Herbal Teas	CCF tea, dosha teas	~30
Golden Milk	Turmeric blends	~15
Digestive Drinks	Churnas, digestive aids	~15

Kitchen & Tools:

Subcategory	Examples	Count
Dinacharya Tools	Tongue scrapers, neti pots	~20
Cookware	Ayurvedic cooking tools	~15
Storage	Containers, dispensers	~10

Books & Resources:

Subcategory	Examples	Count
Guides	Ayurvedic reference books	~20
Cookbooks	Dosha-based recipes	~10
Courses	Educational materials	~10

2.3 Ayurvedic Properties Reference

Rasa (Taste) Effects:

Taste	Sanskrit	Dosha Effect	Examples
Sweet	Madhura	↓V ↓P ↑K	Ashwagandha, licorice, ghee
Sour	Amla	↓V ↑P ↑K	Amalaki, tamarind, citrus
Salty	Lavana	↓V ↑P ↑K	Rock salt, seaweed
Pungent	Katu	↑V ↑P ↓K	Ginger, black pepper, cayenne
Bitter	Tikta	↑V ↓P ↓K	Neem, turmeric, guduchi
Astringent	Kashaya	↑V ↓P ↓K	Triphala, haritaki, pomegranate

Virya (Potency) Effects:

Potency	Sanskrit	Effect	Constitution Notes
Heating	Ushna	Increases Pitta, decreases Vata/Kapha	Good for Vata/Kapha, caution Pitta
Cooling	Shita	Decreases Pitta, may increase Vata/Kapha	Good for Pitta, caution Vata/Kapha

Common Ayurvedic Actions:

Action	Sanskrit	Description	Example Herbs
Rejuvenation	Rasayana	Anti-aging, vitality building	Ashwagandha, Shatavari, Chyawanprash
Digestive Stimulant	Deepana	Kindles digestive fire	Ginger, black pepper, cumin
Mind & Memory	Medhya	Cognitive support	Brahmi, Shankhpushpi, Gotu Kola
Sleep Support	Nidrajanana	Promotes restful sleep	Ashwagandha, Tagara, Jatamansi
Strength Building	Balya	Builds physical strength	Ashwagandha, Shatavari, Bala
Immunity Building	Ojas	Builds vital essence	Chyawanprash, Amalaki, Ghee
Vata Calming	Vatahara	Reduces Vata	Ashwagandha, sesame oil, ghee
Pitta Cooling	Pittahara	Reduces Pitta	Shatavari, aloe, coconut
Kapha Reducing	Kaphahara	Reduces Kapha	Trikatu, honey, ginger

2.4 Product Match Scoring System

Match Indicators:

Match Level	Score	Badge	Description
Excellent	85-100	✅ EXCELLENT	Perfect for constitution and current needs
Good	70-84	✅ GOOD	Beneficial with minor considerations

Moderate	50-69	⚠️ MODERATE	May help but not optimal match
Caution	30-49	⚠️ CAUTION	Use with guidance
Not Recommended	0-29	❌ NOT RECOMMENDED	May aggravate current state

Scoring Factors:

Factor	Weight	Description
Constitutional Match	25%	Does it balance their Prakriti?
Current State Match	25%	Does it address Vikriti (imbalances)?
Protocol Alignment	20%	Is it part of active Projects?
Seasonal Appropriateness	15%	Right for current Ritu?
Health Metric Alignment	10%	Supports CHI, NSL improvement?
Interaction Safety	5%	Any conflicts with current stack?

2.5 Design Philosophy

Twin-Guided Shopping The Digital Twin explains WHY each product is recommended, not just that it's recommended. Users understand how products fit their unique health picture.

Ayurvedic Context Every product shows traditional Ayurvedic properties — Rasa (taste), Virya (potency), Vipaka (post-digestive effect), dosha effects, and traditional uses. This education-commerce blend builds understanding.

Protocol Integration Products from active Projects/Protocols are prominently featured. If someone is on a Nidra Protocol, sleep-supporting products appear first with clear protocol connection.

Seasonal Relevance Recommendations adjust based on Ritucharya (seasonal regimen). Winter shows warming products; summer shows cooling products. This prevents inappropriate purchases.

Constitutional Matching Dosing and selection personalized to Prakriti/Vikriti. A Pitta person gets different recommendations than a Vata person, even for the same health goal.

Quality Transparency Manufacturing standards, testing benchmarks, certifications, and brand reputation clearly displayed. Trust through transparency.

3. User Journey Breakdown

3.1 Main Shop View

Context: User opens Shop section.

Twin Message:

"Based on your elevated Vata and active Nidra Protocol, I've curated products that will support your sleep and calm your nervous system. It's also Shishira (late winter) — a good time for warming, nourishing supplements."

Tab Navigation:

- **For You** — Personalized recommendations
- **Protocols** — Products from active Projects
- **Seasonal** — Current season picks
- **Browse All** — Full category browser

3.2 "What Your Body Needs Today" Section

Context: Personalized product recommendations based on health data.

Section Display:

```
┌──────────────────────────────────────────────────┐
│ 🎯 WHAT YOUR BODY NEEDS TODAY                    │
│ Based on your NSL, CHI, and current state        │
│ ────────────────────────────────────────────────│
│                                                  │
│ Your Focus Areas:                                │
│                                                  │
│ 🌙 Sleep Support (NSL: 68 → improving)           │
│ Your Nidra Protocol needs these essentials       │
│                                                  │
│ 💨 Vata Calming (Elevated 40%)                   │
│ Grounding herbs and warming oils                 │
│                                                  │
│ ❄️ Winter Support (Shishira Ritu)                │
│ Immune-boosting and warming supplements          │
│                                                  │
│ [💭 Why does my body need these?]                │
└──────────────────────────────────────────────────┘
```

Focus Areas Determination:

- Current imbalances (Vikriti)
- Active protocols/projects
- Health metrics needing improvement
- Current season
- Recent health changes

3.3 "From Your Protocols" Section

Context: Products recommended in active Projects.

Section Display:

```
📋 FROM YOUR ACTIVE PROTOCOLS
Products recommended in your current journeys
──────────────────────────────────────────────

🌙 Nidra Optimization Protocol

 Ashwa-      Sesame      Brahmi
 gandha      Oil         Caps
 $24.99      $18.99      $22.99
 ⭐ In       ○ Not in    ○ Not in
 Stack       Stack       Stack

[View all protocol products →]
```

Protocol Product Display:

- Grouped by active project
- Shows "In Stack" status
- Highlights products not yet purchased
- Direct connection to protocol requirements

3.4 Seasonal Recommendations Section

Context: Products appropriate for current season.

Twin Message:

"Winter increases Vata (which you already have elevated) and accumulates Kapha. These products help you stay balanced."

Section Display:

```
❄️ WINTER ESSENTIALS (Shishira Ritu)
Products to thrive in the cold, dry season
──────────────────────────────────────────────

 Warming     Chyawan-    Sesame
 Spice       prash       Oil
 Blend                   (Abhyanga)
 $15.99      $28.99      $18.99
```

```
[View all winter products →]
```

Seasonal Logic:

- Current season determined by Ayurvedic calendar
- Products matched to season's qualities
- Personalized based on constitution (winter affects Vata person differently than Kapha person)

3.5 Shop by Ayurvedic Action

Context: Browse products by traditional Ayurvedic action.

Action Categories:

```
🌿 SHOP BY AYURVEDIC ACTION

┌──────────┐ ┌──────────┐ ┌──────────┐
│ Rasayana │ │ Deepana  │ │ Medhya   │
│ Rejuven- │ │ Digestive│ │ Mind &   │
│ ation    │ │ Stimul.  │ │ Memory   │
└──────────┘ └──────────┘ └──────────┘

┌──────────┐ ┌──────────┐ ┌──────────┐
│ Nidra    │ │ Balya    │ │ Ojas     │
│ Sleep    │ │ Strength │ │ Immunity │
│ Support  │ │ Building │ │ Building │
└──────────┘ └──────────┘ └──────────┘

┌──────────┐ ┌──────────┐ ┌──────────┐
│ Vata     │ │ Pitta    │ │ Kapha    │
│ Balancing│ │ Cooling  │ │ Reducing │
└──────────┘ └──────────┘ └──────────┘
```

3.6 Enhanced Product Card

Context: Product card in any listing.

Card Display:

```
[Product Image]                          ♡   ⊕

ORGANIC INDIA
Ashwagandha
```

```
⭐ 4.8  •  Vata ↓ Kapha ↓

 🎯 Perfect for you
 Calms Vata, supports sleep

$24.99  •  $0.28/serving

[Add to Stack]
```

Card Elements:

- Product image with wishlist/quick-add icons
- Brand name
- Product name
- Rating + Dosha effects (↓ = decreases, ↑ = increases)
- Personalized match tag with brief reason
- Price + price per serving
- Add to Stack CTA

Match Badge Logic:

- "Perfect for you" (85-100 score)
- "Good for you" (70-84 score)
- "May help" (50-69 score)
- "Use with guidance" (30-49 score)
- No badge or warning (<30 score)

3.7 Product Detail View (Supplement/Herb)

Context: User views Ashwagandha detail page.

"Is This Good For You?" Section:

```
🎯 IS THIS GOOD FOR YOU?
──────────────────────────────────────

✅ EXCELLENT MATCH

Your health data shows:
• Elevated Vata (40% vs 25% baseline)
• Active Nidra Protocol
• Sleep onset taking 42+ minutes
• Signs of stress in HRV patterns

Ashwagandha directly addresses these:
```

```
• Calms Vata and nervous system
• Supports healthy cortisol levels
• Promotes restful sleep
• Builds Ojas (vitality reserve)

[💭 How does this fit my existing stack?]
```

Ayurvedic Properties Section:

```
🌿 AYURVEDIC PROPERTIES
──────────────────────────────────────

Sanskrit: Ashwagandha (अश्वगंधा)
Meaning: "Smell of horse" - gives strength
```

Rasa (Taste)	Tikta (Bitter), Kashaya (Astringent)
Virya (Potency)	Ushna (Heating)
Vipaka (Post-dig)	Madhura (Sweet)
Dosha Effect	Vata ↓ Kapha ↓ (↑ Pitta in excess)
Actions	Rasayana, Balya, Nidrajanana, Medhya

```
Traditional Uses:
• Stress adaptation (Adaptogen)
• Sleep support (Nidrajanana)
• Strength building (Balya)
• Rejuvenation (Rasayana)
• Cognitive support (Medhya)

[📖 Learn more about Ashwagandha]
```

Personalized Dosage Section:

```
⏰ DOSAGE & TIMING
Personalized for your constitution
```

```
Recommended for Pitta-Vata (You):

Dosage: 1 capsule (300mg)
Frequency: Once daily
Best Time: Evening, 1 hour before bed
With: Warm milk or water

⚠️ Note: Ashwagandha is warming. As a Pitta-
Vata, monitor for any signs of excess heat
(acidity, irritability). Take with cooling
agents like milk if needed.

Duration: Minimum 4-6 weeks for full effect
```

Quality Indicators:

```
✅ QUALITY INDICATORS

Inactive Ingredients      SAFE          ▼
Manufacturing Standards   HIGH          ▼
Testing Benchmarks        HIGH          ▼
Product Quality           HIGH          ▼
Product Certifications    USDA Organic  ▼
Brand Reputation          Excellent     ▼
```

3.8 Product Detail View (Food)

Context: User views Blueberries detail page.

"Is This Good For You?" Section:

```
🎯 IS THIS GOOD FOR YOU?
```

```
✅ GOOD MATCH

Your health data shows:
• Slightly elevated Pitta tendency
• Need for antioxidant support
• CHI could use improvement

Blueberries help because:
• Sweet taste cools Pitta
• Rich in antioxidants for recovery
• Supports brain health (Medhya)
• Light and easy to digest

⚠️ Note: Best eaten at room temperature (not
cold from fridge) to support Agni.

[💭 When should I eat these?]
```

"Best Way to Eat" Section (Food-specific):

```
⏰ BEST WAY TO EAT
Personalized for your constitution
──────────────────────────────────────

For Pitta-Vata (You):

Best Time: Morning or mid-morning snack
Avoid: Evening (fruit best before 4 PM)
Preparation: Room temperature, not cold
Combine With: Alone or with nuts
Avoid With: Dairy, other fruits (combining)

Seasonal Note:
In winter (now), berries are cooling - eat in
moderation and at room temperature.
```

3.9 My Stack (Cart) View

Context: User views their shopping cart.

Twin Message:

"Your current stack looks well-balanced for your Nidra Protocol. I notice you don't have Brahmi yet — it pairs well with Ashwagandha for sleep and cognition."

Stack Display:

```
🛒 YOUR STACK (3 items)

  [img] Ashwagandha - Organic India
        $24.99  •  90 caps
        For: Nidra Protocol ✓
        Timing: Evening, before bed
                                [-] 1 [+]

  [img] Sesame Oil - Banyan Botanicals
        $18.99  •  16 oz
        For: Abhyanga (self-massage)
        Timing: Morning, before shower
                                [-] 1 [+]

  [img] CCF Tea - Organic
        $12.99  •  100g
        For: Digestive support
        Timing: After meals
                                [-] 1 [+]
```

Twin Suggestion:

```
💡 TWIN SUGGESTS ADDING

  [img] Brahmi - Cognitive Support
        $22.99
        Why: Complements Ashwagandha for
        sleep & calms racing mind
        [Add to Stack]
```

Supplement Schedule Preview:

```
📅 YOUR SUPPLEMENT SCHEDULE
Once purchased, I'll help you schedule:

Morning: Sesame oil (Abhyanga)
After meals: CCF Tea
Evening: Ashwagandha (1hr before bed)
```

3.10 Search and Filtering

Context: User searches or filters products.

Search Features:

- Natural language search ("herbs for sleep")
- Product name search
- Ingredient search
- Ayurvedic term search ("Rasayana")

Filter Options:

- Category
- Dosha effect (reduces Vata, reduces Pitta, reduces Kapha)
- Ayurvedic action
- Price range
- Brand
- Certifications (Organic, Non-GMO, etc.)
- Match score (Excellent, Good, Moderate)
- Dietary (Vegan, Vegetarian, Gluten-free)

3.11 Post-Purchase Integration

Context: User completes purchase.

Integration Flow:

```
Purchase Complete
      │
      ▼
Products added to Supplements section
      │
      ▼
Twin creates dosing schedule
      │
      ▼
Reminders added to Today view
      │
      ▼
```

Tracking integrated with Health metrics

Twin Message After Purchase:

"Your order is confirmed! I've added your new supplements to your stack. Starting tomorrow, I'll remind you:

- Morning: Sesame oil for Abhyanga
- After meals: CCF Tea
- Evening (1hr before bed): Ashwagandha with warm milk

Shall I set up these reminders?"

4. Technical Architecture

4.1 High-Level Architecture

```
┌──────────────
```

```
│   │  Products Store  │   │  Cart Store      │   │  Shop UI
Store   │   │
│   │                   │   │                  │   │
│   │
│   │  • Product list   │   │ • Stack items    │   │ • Current
view   │   │
│   │  • Categories     │   │ • Wishlist       │   │ • Filters
│   │
│   │  • Recommendations│   │ • Suggestions    │   │ • Search
│   │
│   │  • Match scores   │   │ • Orders         │   │ • Messages
│   │
│   └───────────────────┘   └──────────────────┘
└───────────────────┘   │
└──────────────────────────────────────────────────────────────────
──────────┘
                                    │
                                    ▼
┌──────────────────────────────────────────────────────────────────
──────────┐
│                              SERVICE LAYER
│
│   ┌──────────────────────┐                          │
┌──────────────────────────────────┐    │
│   │   Shop Agent         │   │     Product Matching Service
│   │
│   │   (Claude API)       │   │
│   │
│   │                      │   │  • Match score calculation
│   │
│   │  • Recommendations   │   │  • Constitutional filtering
│   │
│   │  • Product guidance  │   │  • Seasonal filtering
│   │
│   │  • Personalization   │   │  • Protocol alignment
│   │
│   │  • Stack analysis    │   │  • Interaction checking
│   │
│   └──────────────────────┘
└──────────────────────────────────┘    │
│
│
│   ┌──────────────────────┐
┌──────────────────────────────────┐    │
```

```
│   │  Dosing Service      │  │      Integration Service
│   │                      │  │
│   │                      │  │
│   │  • Constitutional    │  │  • Supplements section sync
│   │                      │  │
│   │    dosing            │  │  • Today section tasks
│   │                      │  │
│   │  • Timing recommend  │  │  • Protocol connections
│   │                      │  │
│   │  • Interaction check │  │  • Health tracking
│   │                      │  │
│   └──────────────────────┘  │
└─────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────
──────────┘
                                    │
                                    ▼
┌──────────────────────────────────────────────────────────────────────────────
──────────┐
│                                DATA LAYER
│
│   ┌──────────────────────┐  ┌──────────────────────────────┐     │
│   │   Local Cache        │  │      Backend APIs
│   │                      │  │
│   │                      │  │
│   │  • Product catalog   │  │  • Product service
│   │                      │  │
│   │  • Match scores      │  │  • Order service
│   │                      │  │
│   │  • Cart state        │  │  • Recommendation service
│   │                      │  │
│   │  • Wishlist          │  │  • Inventory service
│   │                      │  │
│   └──────────────────────┘  │
└─────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────
──────────┘
```

4.2 Data Flow: Loading Shop Main View

```
User opens Shop section
        │
        ▼
┌──────────────────────────┐
```

```
┌──────────────────────────┐
│ Get user context:        │ ── Constitution, imbalances,
protocols
│  • Prakriti              │
│  • Vikriti               │
│  • Active projects       │
│  • Current supplements   │
│  • Health metrics        │
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Get current season       │ ── Ritucharya context
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Fetch personalized       │ ── AI-scored recommendations
│ recommendations          │
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Fetch protocol           │ ── Products from active Projects
│ products                 │
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Fetch seasonal           │ ── Current season products
│ products                 │
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Claude generates         │ ── Contextual guidance
│ Twin message             │
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Render shop view         │
│ with all sections        │
└──────────────────────────┘
```

4.3 Data Flow: Product Match Scoring

```
Product needs scoring for user
```

```
        │
        ▼
┌──────────────────────────┐
│ Get product Ayurvedic    │ ── Rasa, Virya, Vipaka, dosha
effects
│ properties               │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Get user context:        │
│ • Prakriti (25%)         │
│ • Vikriti (25%)          │
│ • Active protocols       │
│ • Season (15%)           │
│ • Health metrics (10%)   │
│ • Current stack (5%)     │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Calculate component      │
│ scores:                  │
│                          │
│ 1. Constitutional        │ ── Does it balance Prakriti?
│    match                 │
│                          │
│ 2. Current state         │ ── Does it address Vikriti?
│    match                 │
│                          │
│ 3. Protocol              │ ── Is it in active projects?
│    alignment             │
│                          │
│ 4. Seasonal              │ ── Right for current Ritu?
│    appropriateness       │
│                          │
│ 5. Health metric         │ ── Supports CHI/NSL?
│    alignment             │
│                          │
│ 6. Interaction           │ ── Safe with current stack?
│    safety                │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
```

```
│ Calculate weighted     │ ── Sum of weighted component
scores
│ final score (0-100)    │
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Generate match badge   │ ── Excellent/Good/Moderate/Caution
│ and reason             │
└────────────────────────┘
```

4.4 Data Flow: Personalized Dosing

```
Generate personalized dosing for product
        │
        ▼
┌────────────────────────┐
│ Get product standard   │ ── Base dosing information
│ dosing info            │
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Get user profile:      │
│ • Prakriti             │
│ • Current Vikriti      │
│ • Age/weight (if       │
│   available)           │
│ • Current supplements  │
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Adjust dosing:         │
│ • Vata: smaller doses, │
│   more frequently      │
│ • Pitta: moderate,     │
│   with cooling         │
│ • Kapha: larger doses, │
│   less frequently      │
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Determine timing:      │
│ • Based on product     │
│   action               │
```

```
| • Based on user's      |
|   routine              |
| • Based on protocol    |
|   needs                |
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Add constitutional     │ ── Specific notes for user's type
│ warnings/notes         │
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Check interactions     │ ── Against current stack
└────────────────────────┘
```

4.5 Data Flow: Post-Purchase Integration

```
Order completed
        │
        ▼
┌────────────────────────┐
│ Get ordered products   │ ── With personalized dosing
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Create supplement      │ ── Add to Supplements section
│ entries                │
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Generate dosing        │ ── Based on all supplements
│ schedule               │
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Create Today tasks     │ ── Reminder cards
└────────────────────────┘
        │
        ▼
┌────────────────────────┐
│ Link to protocols      │ ── If product is part of protocol
│ (if applicable)        │
```

```
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Set up tracking          │ ── For health metrics correlation
└──────────────────────────┘
```

4.6 Product Match Score Algorithm

```
MATCH SCORE CALCULATION:

Inputs:
- Product: Ayurvedic properties (rasa, virya, vipaka,
dosha_effects, actions)
- User: Prakriti, Vikriti, active_protocols, current_season,
health_metrics, current_stack

Components:

1. CONSTITUTIONAL MATCH (25%)
   For each elevated dosha in Prakriti:
     If product reduces that dosha: +points
     If product increases that dosha: -points
   Score based on how well it balances constitution

2. CURRENT STATE MATCH (25%)
   For each elevated dosha in Vikriti (current imbalances):
     If product reduces that dosha: +points (more weight than
Prakriti)
     If product increases that dosha: -points (more weight)
   Score based on how well it addresses current issues

3. PROTOCOL ALIGNMENT (20%)
   If product is explicitly recommended in active protocol:
+100%
   If product category matches protocol focus: +50%
   If product action aligns with protocol goals: +25%
   No protocol connection: +0%

4. SEASONAL APPROPRIATENESS (15%)
   Check product's heating/cooling nature vs. season needs:
     Winter: Heating products score higher
     Summer: Cooling products score higher
   Check if product counters seasonal dosha accumulation

5. HEALTH METRIC ALIGNMENT (10%)
```

```
   If product action relates to user's weak health metrics:
     Sleep support + low NSL: +points
     Digestive support + low Agni score: +points
     Etc.

6. INTERACTION SAFETY (5%)
   Check for known interactions with current stack
   Check for contraindications based on health data
   Deduct points for any concerns

Final Score = Weighted sum of components (0-100)

Match Level:
- 85-100: EXCELLENT MATCH
- 70-84: GOOD MATCH
- 50-69: MODERATE MATCH
- 30-49: USE WITH CAUTION
- 0-29: NOT RECOMMENDED
```

5. AI Integration & Prompting Strategy

5.1 The Role of Claude in Shop Section

Claude serves as a shopping guide and product advisor:

Capability	Description
Personalized Recommendations	Curate products based on health profile
Product Assessment	Explain why a product is/isn't good for user
Dosing Guidance	Provide constitutional dosing recommendations
Stack Analysis	Analyze how products work together
Q&A	Answer questions about products
Ayurvedic Education	Explain traditional properties and uses

5.2 System Prompt for Shop Section

```
You are the Ayurvedic Digital Twin for AyuAware, serving as a
knowledgeable shopping guide and product advisor. Your role
is to
help users find products that truly match their constitution,
current health needs, and active protocols.

## YOUR ROLE IN THIS SECTION
```

You are a trusted advisor. You help users understand WHY products
are or aren't suitable for them, explain Ayurvedic properties in
accessible terms, and ensure purchases support their health journey.

AYURVEDIC PRODUCT KNOWLEDGE

Product Properties:
- Rasa (Taste): Sweet, Sour, Salty, Pungent, Bitter, Astringent
- Virya (Potency): Heating (Ushna) or Cooling (Shita)
- Vipaka (Post-digestive): Sweet, Sour, or Pungent
- Dosha Effects: Which doshas it increases/decreases
- Actions: Rasayana, Deepana, Medhya, Nidra, Balya, etc.

Constitutional Considerations:
- Vata types: Need grounding, warming, nourishing products
- Pitta types: Need cooling, calming, moderate products
- Kapha types: Need stimulating, warming, lightening products

Dosing Principles:
- Vata: Smaller doses, more frequently, with warmth
- Pitta: Moderate doses, with cooling agents
- Kapha: Larger doses, less frequently, with stimulation

Seasonal Considerations:
- Winter (Shishira/Hemanta): Warming, nourishing, Vata-calming
- Spring (Vasanta): Detoxifying, Kapha-reducing
- Summer (Grishma): Cooling, Pitta-calming, lighter
- Monsoon (Varsha): Digestive support, warming
- Autumn (Sharad): Pitta-clearing, grounding

COMMUNICATION APPROACH

When recommending products:
- Connect to user's specific health picture
- Explain Ayurvedic reasoning
- Note any considerations or cautions
- Suggest complementary products

When explaining match scores:

```
- Be specific about why something matches or doesn't
- Reference their actual health data
- Provide actionable guidance

When discussing dosing:
- Personalize to their constitution
- Include timing recommendations
- Note any constitutional warnings

When answering questions:
- Be informative and educational
- Connect to their specific situation
- Suggest related products if helpful

## WHAT NOT TO DO

- Don't recommend products that aggravate their current
imbalances
- Don't ignore constitutional considerations
- Don't provide generic recommendations without
personalization
- Don't recommend too many products at once
- Don't ignore potential interactions
- Don't make medical claims or diagnose conditions
- Don't recommend against user's active protocols
```

5.3 Context Building for Shop Section

Every Claude call includes:

```
CONTEXT PACKAGE STRUCTURE:

## User Profile
- Name: {user_name}
- Prakriti: {prakriti_percentages}
- Primary constitution: {constitution_type}
- Current Vikriti: {vikriti_percentages}
- Current imbalances: {imbalance_list}
- Health scores: {CHI, NSL, Agni, Ojas}

## Current Stack
- Active supplements: {supplement_list_with_dosing}
- Recent purchases: {recent_order_list}

## Active Protocols
- Protocol name: {protocol_name}
```

```
- Protocol focus: {focus_area}
- Recommended products: {product_list}
- Products already purchased: {purchased_list}

## Seasonal Context
- Current season: {ritu_name}
- Seasonal qualities: {guna_list}
- Seasonal dosha accumulation: {accumulating_dosha}

## Shopping Context
- Current product (if viewing): {product_details}
- Cart contents: {cart_items}
- Search/filter context: {query_context}

## Specific Request
- What user is asking/doing: {query_context}
```

5.4 Prompt Templates by Scenario

Shop Welcome Message:

```
Generate a welcome message for the Shop section.

{CONTEXT_PACKAGE}

Generate a message that:

1. **Acknowledges their current health picture**
   - Current imbalances
   - Active protocols

2. **Sets shopping context**
   - What types of products would help them
   - Seasonal relevance

3. **Creates helpful framing**
   - Brief, actionable guidance

Tone: Helpful advisor, not salesperson
Length: 3-4 sentences
```

Product Assessment:

```
Assess whether this product is good for this user.

{CONTEXT_PACKAGE}

Product:
```

- Name: {product_name}
- Ayurvedic Properties: {rasa, virya, vipaka, dosha_effects}
- Actions: {ayurvedic_actions}
- Match Score: {calculated_score}

Generate an assessment that:

1. **States the match level clearly**
   - Excellent/Good/Moderate/Caution

2. **Explains why based on their health data**
   - Reference specific imbalances
   - Reference protocols if relevant
   - Reference season if relevant

3. **Lists how this product helps them**
   - Specific benefits for their situation

4. **Notes any considerations**
   - Constitutional warnings
   - Timing considerations
   - Interactions if any

Tone: Informative, personalized
Length: Appropriate to complexity

Personalized Dosing:

Generate personalized dosing recommendations.

{CONTEXT_PACKAGE}

Product:
- Name: {product_name}
- Standard Dosing: {standard_dose}
- Ayurvedic Properties: {properties}

Generate dosing guidance that:

1. **Provides specific dose**
   - Amount
   - Frequency

2. **Recommends timing**
   - Best time of day
   - Relation to meals

```
3. **Suggests anupana (vehicle)**
   - What to take it with
   - Constitutional considerations

4. **Adds any warnings**
   - Constitutional notes
   - What to watch for

5. **Notes duration**
   - How long to take
   - When to expect effects

Tone: Clear, practical guidance
```

Stack Analysis:

```
Analyze this user's current/proposed supplement stack.

{CONTEXT_PACKAGE}

Current Stack: {current_supplements}
Proposed Additions: {new_products}

Generate analysis that:

1. **Assesses overall stack balance**
   - Does it address their needs?
   - Any gaps?

2. **Checks for interactions**
   - Any concerns?
   - Timing considerations

3. **Suggests improvements**
   - Missing products
   - Potential removals
   - Dosing adjustments

4. **Creates schedule**
   - Morning products
   - Midday products
   - Evening products
   - With meals vs. without

Tone: Comprehensive, helpful
```

Product Q&A:

```
Answer this question about a product.

{CONTEXT_PACKAGE}

Product: {product_details}
Question: {user_question}

Generate an answer that:

1. **Directly addresses the question**
2. **Personalizes to their situation**
3. **Includes relevant Ayurvedic context**
4. **Suggests next steps if helpful**

Tone: Educational, personalized
Length: Appropriate to question complexity
```

6. Data Models & State Management

6.1 Core Data Types

Product:

```
Product:
  - id: string
  - name: string
  - brand: string
  - category: ProductCategory
  - subcategory: string
  - description: string
  - shortDescription: string
  - images: ProductImage[]
  - price: number
  - currency: string
  - pricePerServing: number | null
  - servingsPerContainer: number | null
  - format: string (capsule, powder, oil, etc.)
  - size: string
  - inStock: boolean
  - stockLevel: number
  - rating: number
  - reviewCount: number
  - ayurvedicProperties: AyurvedicProperties
  - ingredients: Ingredient[]
  - nutritionFacts: NutritionFacts | null
```

```
  - qualityIndicators: QualityIndicators
  - certifications: string[]
  - tags: string[]
  - relatedProducts: string[]
  - createdAt: Date
  - updatedAt: Date

ProductCategory:
  - 'supplements_herbs' | 'foods_ingredients' |
'oils_body_care' |
    'teas_beverages' | 'kitchen_tools' | 'books_resources'

ProductImage:
  - url: string
  - alt: string
  - isPrimary: boolean
```

AyurvedicProperties:

```
AyurvedicProperties:
  - sanskritName: string | null
  - meaning: string | null
  - rasa: Rasa[] (tastes)
  - virya: 'ushna' | 'shita' (heating/cooling)
  - vipaka: 'madhura' | 'amla' | 'katu' (post-digestive)
  - doshaEffects: DoshaEffects
  - gunas: string[] (qualities)
  - actions: AyurvedicAction[]
  - traditionalUses: string[]
  - contraindications: string[]
  - tissueAffinity: string[] (dhatus affected)
  - channelAffinity: string[] (srotas affected)

Rasa:
  - 'madhura' | 'amla' | 'lavana' | 'katu' | 'tikta' |
'kashaya'
  (sweet, sour, salty, pungent, bitter, astringent)

DoshaEffects:
  - vata: 'increases' | 'decreases' | 'neutral'
  - pitta: 'increases' | 'decreases' | 'neutral'
  - kapha: 'increases' | 'decreases' | 'neutral'
  - notes: string | null

AyurvedicAction:
  - action: string (Rasayana, Deepana, Medhya, etc.)
```

```
  - description: string
```

QualityIndicators:

```
QualityIndicators:
  - inactiveIngredients: QualityLevel
  - manufacturingStandards: QualityLevel
  - testingBenchmarks: QualityLevel
  - productQuality: QualityLevel
  - certifications: string[]
  - brandReputation: QualityLevel
  - details: QualityDetail[]

QualityLevel:
  - 'excellent' | 'high' | 'good' | 'moderate' | 'low' |
'unknown'

QualityDetail:
  - category: string
  - level: QualityLevel
  - description: string
  - expandedInfo: string
```

ProductMatch:

```
ProductMatch:
  - productId: string
  - userId: string
  - score: number (0-100)
  - matchLevel: 'excellent' | 'good' | 'moderate' | 'caution'
| 'not_recommended'
  - components: MatchComponents
  - reason: string (brief personalized reason)
  - benefits: string[]
  - considerations: string[]
  - calculatedAt: Date

MatchComponents:
  - constitutionalMatch: number (0-100)
  - currentStateMatch: number (0-100)
  - protocolAlignment: number (0-100)
  - seasonalAppropriateness: number (0-100)
  - healthMetricAlignment: number (0-100)
  - interactionSafety: number (0-100)
```

PersonalizedDosing:

```
PersonalizedDosing:
  - productId: string
```

```
  - userId: string
  - dosage: string
  - frequency: string
  - timing: string
  - bestTime: string
  - withFood: boolean
  - anupana: string (what to take with)
  - duration: string
  - constitutionalNotes: string
  - warnings: string[]
  - generatedAt: Date
```

CartItem:

```
CartItem:
  - productId: string
  - product: Product
  - quantity: number
  - matchScore: number
  - personalizedDosing: PersonalizedDosing | null
  - protocolConnection: string | null
  - addedAt: Date

Cart:
  - items: CartItem[]
  - subtotal: number
  - shipping: number
  - tax: number
  - total: number
  - twinSuggestions: ProductSuggestion[]
  - schedulePreview: SchedulePreview
```

ProductSuggestion:

```
ProductSuggestion:
  - productId: string
  - product: Product
  - reason: string
  - matchScore: number
  - suggestionType: 'complement' | 'complete_protocol' |
'seasonal' | 'stack_gap'
```

Order:

```
Order:
  - id: string
  - userId: string
  - items: OrderItem[]
  - status: OrderStatus
```

```
  - subtotal: number
  - shipping: number
  - tax: number
  - total: number
  - shippingAddress: Address
  - paymentMethod: PaymentMethod
  - trackingNumber: string | null
  - estimatedDelivery: Date | null
  - createdAt: Date
  - updatedAt: Date

OrderItem:
  - productId: string
  - productName: string
  - quantity: number
  - price: number
  - personalizedDosing: PersonalizedDosing | null

OrderStatus:
  - 'pending' | 'confirmed' | 'processing' | 'shipped' |
    'delivered' | 'cancelled' | 'returned'
```

Wishlist:

```
WishlistItem:
  - productId: string
  - product: Product
  - matchScore: number
  - addedAt: Date
  - priceAtAdd: number
  - notifyOnPriceDrop: boolean
```

6.2 State Structure

```
ShopState:
  # Visit Context
  - isFirstVisit: boolean
  - lastVisit: Date | null

  # Products
  - products: Product[]
  - isLoadingProducts: boolean
  - productsError: Error | null
  - productCache: Record<string, Product>

  # Categories
  - categories: ProductCategory[]
```

```
  - selectedCategory: string | null
  - selectedSubcategory: string | null

  # Match Scores
  - matchScores: Record<string, ProductMatch>
  - isCalculatingMatches: boolean

  # Current Product
  - selectedProduct: Product | null
  - selectedProductMatch: ProductMatch | null
  - selectedProductDosing: PersonalizedDosing | null
  - isLoadingProduct: boolean

  # Recommendations
  - forYouProducts: Product[]
  - protocolProducts: Record<string, Product[]>
  - seasonalProducts: Product[]
  - isLoadingRecommendations: boolean

  # Search & Filters
  - searchQuery: string
  - searchResults: Product[]
  - filters: ShopFilters
  - sortBy: SortOption

  # Cart
  - cart: Cart
  - isLoadingCart: boolean
  - cartError: Error | null

  # Wishlist
  - wishlist: WishlistItem[]
  - isLoadingWishlist: boolean

  # Orders
  - orders: Order[]
  - currentOrder: Order | null
  - isLoadingOrders: boolean

  # UI State
  - activeTab: 'for_you' | 'protocols' | 'seasonal' |
'browse'
  - viewMode: 'grid' | 'list'
  - filterPanelOpen: boolean
  - scrollPosition: number
```

```
  # Chat
  - messages: ShopMessage[]
  - isTyping: boolean

ShopFilters:
  - category: string | null
  - doshaEffect: string[] (reduces_vata, reduces_pitta,
reduces_kapha)
  - ayurvedicAction: string[]
  - priceRange: [number, number] | null
  - brand: string[]
  - certifications: string[]
  - matchLevel: string[] (excellent, good, moderate)
  - dietary: string[] (vegan, vegetarian, gluten_free)
  - inStock: boolean
```

6.3 State Management Approach

Zustand Stores:

```
Stores:

1. productsStore
   - Product catalog
   - Categories
   - Search
   - Filters
   - Match scores

2. cartStore
   - Cart items
   - Cart operations
   - Suggestions
   - Schedule preview

3. wishlistStore
   - Wishlist items
   - Price tracking
   - Notifications

4. ordersStore
   - Order history
   - Current order
   - Tracking
```

```
5. shopUIStore
   - View states
   - Tab navigation
   - Messages
   - Modals
```

7. Component Requirements

7.1 Main Shop Components

ShopHeader

Header with Twin message and navigation:

- Twin avatar with contextual message
- Search bar
- Cart icon with count
- Tab navigation (For You, Protocols, Seasonal, Browse All)

WhatYourBodyNeedsCard

Personalized focus areas:

- Focus area list with icons
- Health metric context
- "Why does my body need these?" link
- Connection to recommendations

ProtocolProductsSection

Products from active protocols:

- Protocol name header
- Product grid
- "In Stack" status badges
- View all link

SeasonalPicksSection

Current season products:

- Season name and description
- Twin explanation
- Product grid
- View all link

AyurvedicActionGrid

Browse by action:

- Action category icons
- Category names (Sanskrit + English)
- Tap to filter

CategoryBrowser

Full category list:

- Category cards with icons
- Product counts
- Subcategory expansion

7.2 Product Card Components

ProductCard

Product in listing:

- Product image
- Wishlist/quick-add icons
- Brand name
- Product name
- Rating + dosha effects
- Match badge with brief reason
- Price + price per serving
- Add to Stack button

ProductCardCompact

Smaller card for grids:

- Image
- Name
- Price
- Match indicator

MatchBadge

Personalized match indicator:

- Match level (Excellent/Good/etc.)
- Brief reason
- Color-coded

7.3 Product Detail Components

ProductImageGallery

Image viewer:

- Main image
- Thumbnail navigation
- Zoom capability

IsThisGoodForYouCard

Personalized assessment:

- Match level (prominent)
- "Your health data shows" section
- "This product helps because" section
- Considerations/warnings
- "How does this fit my stack?" link

AyurvedicPropertiesCard

Traditional properties:

- Sanskrit name and meaning
- Properties table (Rasa, Virya, Vipaka, Dosha)
- Actions list
- Traditional uses
- Learn more link

PersonalizedDosingCard

Constitutional dosing:

- Recommended dosage
- Frequency
- Best time
- What to take with (anupana)
- Constitutional warnings
- Duration

QualityIndicatorsCard

Quality transparency:

- Quality metric rows
- Expandable details
- Certifications

RelatedProductsSection

Complementary products:

- "Often bought together"
- Product cards
- Add to stack shortcuts

7.4 Cart Components

CartHeader

Cart overview:

- Twin message about stack
- Item count

CartItemCard

Item in cart:

- Product image
- Product name and details
- Price
- Protocol connection (if any)
- Dosing timing
- Quantity controls
- Remove button

TwinSuggestionsCard

AI recommendations:

- Suggested product
- Reason for suggestion
- Add button

SchedulePreviewCard

Dosing schedule preview:

- Morning items
- Midday items
- Evening items
- "Once purchased..." context

CartSummary

Order summary:

- Subtotal
- Shipping
- Total
- Checkout button

7.5 Search & Filter Components

SearchBar

Search input:

- Search icon
- Input field
- Clear button
- Recent searches

FilterPanel

Filter controls:

- Category dropdown
- Dosha effect checkboxes
- Action checkboxes
- Price range slider
- Brand multi-select
- Certifications
- Match level filter
- Apply/Clear buttons

ActiveFilters

Currently applied filters:

- Filter chips
- Remove individual
- Clear all

SortDropdown

Sort options:

- Relevance (default)

- Price low-high
- Price high-low
- Rating
- Match score

7.6 Animation Requirements

Animation	Duration	Easing	Trigger
Add to cart	400ms	ease-out	Add button tap
Cart icon bounce	300ms	bounce	Item added
Match badge appear	300ms	ease-out	Card loads
Product image zoom	200ms	ease-out	Image tap
Filter panel slide	300ms	ease-out	Filter toggle
Wishlist heart fill	200ms	ease-out	Wishlist tap
Remove item	300ms	ease-out	Remove tap
Price update	200ms	ease-out	Quantity change
Tab switch	200ms	ease-out	Tab tap

8. API Requirements

8.1 REST Endpoints

Get Shop Home:

```
GET /api/v1/shop/home

Response:
{
  "forYouProducts": [
    {
      "id": "prod_001",
      "name": "Ashwagandha",
      "brand": "Organic India",
      "price": 24.99,
      "image": "...",
      "matchScore": 92,
      "matchLevel": "excellent",
      "matchReason": "Calms Vata, supports sleep"
```

```
    },
    ...
  ],
  "protocolProducts": {
    "nidra_protocol": {
      "protocolName": "Nidra Optimization",
      "products": [...],
      "inStackStatus": { "prod_001": true, "prod_002":
false }
    }
  },
  "seasonalProducts": {
    "season": "Shishira",
    "seasonName": "Late Winter",
    "products": [...]
  },
  "focusAreas": [
    {
      "icon": "🌙",
      "title": "Sleep Support",
      "subtitle": "Your Nidra Protocol needs these",
      "metric": "NSL: 68 → improving"
    },
    ...
  ]
}
```

Get Product Detail:

```
GET /api/v1/shop/products/:productId

Response:
{
  "product": {
    "id": "prod_001",
    "name": "Ashwagandha",
    "brand": "Organic India",
    "category": "supplements_herbs",
    "description": "...",
    "images": [...],
    "price": 24.99,
    "pricePerServing": 0.28,
    "servingsPerContainer": 90,
    "format": "Vegetarian capsule",
    "inStock": true,
    "rating": 4.8,
```

```
    "reviewCount": 234,
    "ayurvedicProperties": {
      "sanskritName": "Ashwagandha",
      "meaning": "Smell of horse - gives strength",
      "rasa": ["tikta", "kashaya"],
      "virya": "ushna",
      "vipaka": "madhura",
      "doshaEffects": {
        "vata": "decreases",
        "pitta": "increases_in_excess",
        "kapha": "decreases"
      },
      "actions": [
        { "action": "Rasayana", "description": "Rejuvenation"
},
        { "action": "Balya", "description": "Strength
building" },
        { "action": "Nidrajanana", "description": "Sleep
promoting" }
      ],
      "traditionalUses": [...]
    },
    "qualityIndicators": {...},
    "ingredients": [...],
    "relatedProducts": [...]
  },
  "match": {
    "score": 92,
    "matchLevel": "excellent",
    "components": {...},
    "benefits": [
      "Calms Vata and nervous system",
      "Supports healthy cortisol levels",
      "Promotes restful sleep"
    ],
    "considerations": [
      "Warming - monitor for excess heat as Pitta-Vata"
    ]
  },
  "personalizedDosing": {
    "dosage": "1 capsule (300mg)",
    "frequency": "Once daily",
    "timing": "Evening",
    "bestTime": "1 hour before bed",
    "anupana": "Warm milk or water",
```

```
    "constitutionalNotes": "As a Pitta-Vata, take with
cooling agents...",
    "duration": "Minimum 4-6 weeks for full effect"
  }
}
```

Search Products:

```
GET /api/v1/shop/search?q={query}&filters={filters}

Response:
{
  "query": "sleep herbs",
  "totalResults": 24,
  "products": [
    {
      "id": "prod_001",
      "name": "Ashwagandha",
      "matchScore": 92,
      ...
    },
    ...
  ],
  "filters": {
    "applied": {...},
    "available": {
      "categories": [...],
      "doshaEffects": [...],
      "actions": [...],
      "brands": [...],
      "priceRange": { "min": 8.99, "max": 89.99 }
    }
  },
  "suggestions": [
    "sleep support",
    "nidra herbs"
  ]
}
```

Get Cart:

```
GET /api/v1/shop/cart

Response:
{
  "items": [
    {
      "productId": "prod_001",
```

```
      "product": {...},
      "quantity": 1,
      "matchScore": 92,
      "personalizedDosing": {...},
      "protocolConnection": "Nidra Protocol"
    },
    ...
  ],
  "subtotal": 56.97,
  "shipping": 5.99,
  "tax": 4.56,
  "total": 67.52,
  "twinSuggestions": [
    {
      "productId": "prod_005",
      "product": {...},
      "reason": "Complements Ashwagandha for sleep &
cognition",
      "matchScore": 88
    }
  ],
  "schedulePreview": {
    "morning": ["Sesame oil (Abhyanga)"],
    "afterMeals": ["CCF Tea"],
    "evening": ["Ashwagandha (1hr before bed)"]
  }
}
```


Add to Cart:

```
POST /api/v1/shop/cart/add

Request:
{
  "productId": "prod_001",
  "quantity": 1
}

Response:
{
  "success": true,
  "cart": {...},
  "addedItem": {
    "product": {...},
    "matchScore": 92,
    "personalizedDosing": {...}
```

```
  }
}
```

Update Cart Item:

```
PUT /api/v1/shop/cart/items/:productId

Request:
{
  "quantity": 2
}

Response:
{
  "success": true,
  "cart": {...}
}
```

Remove from Cart:

```
DELETE /api/v1/shop/cart/items/:productId

Response:
{
  "success": true,
  "cart": {...}
}
```

Checkout:

```
POST /api/v1/shop/checkout

Request:
{
  "shippingAddress": {...},
  "paymentMethodId": "pm_001",
  "createSupplementEntries": true
}

Response:
{
  "success": true,
  "order": {
    "id": "order_001",
    "status": "confirmed",
    "total": 67.52,
    "estimatedDelivery": "2026-02-05"
  },
  "supplementsCreated": [
```

```
    { "name": "Ashwagandha", "scheduleCreated": true }
  ],
  "todayTasksCreated": 3
}
```

Get Wishlist:

```
GET /api/v1/shop/wishlist

Response:
{
  "items": [
    {
      "productId": "prod_003",
      "product": {...},
      "matchScore": 85,
      "addedAt": "2026-01-25",
      "priceAtAdd": 28.99,
      "currentPrice": 24.99,
      "priceDropped": true
    },
    ...
  ]
}
```

Add to Wishlist:

```
POST /api/v1/shop/wishlist/add

Request:
{
  "productId": "prod_003"
}

Response:
{
  "success": true,
  "wishlist": {...}
}
```

Get Orders:

```
GET /api/v1/shop/orders

Response:
{
  "orders": [
    {
      "id": "order_001",
```

```
      "status": "delivered",
      "total": 67.52,
      "itemCount": 3,
      "createdAt": "2026-01-20",
      "deliveredAt": "2026-01-25"
    },
    ...
  ]
}
```

Get Order Detail:

```
GET /api/v1/shop/orders/:orderId

Response:
{
  "order": {
    "id": "order_001",
    "items": [...],
    "status": "delivered",
    "subtotal": 56.97,
    "shipping": 5.99,
    "tax": 4.56,
    "total": 67.52,
    "shippingAddress": {...},
    "trackingNumber": "1Z999...",
    "timeline": [
      { "status": "confirmed", "date": "2026-01-20" },
      { "status": "shipped", "date": "2026-01-22" },
      { "status": "delivered", "date": "2026-01-25" }
    ]
  }
}
```

8.2 WebSocket Events (Optional)

```
WebSocket Events:

shop.price_drop
{
  "productId": "prod_003",
  "productName": "Chyawanprash",
  "oldPrice": 28.99,
  "newPrice": 24.99,
  "inWishlist": true
}
```

```
shop.back_in_stock
{
  "productId": "prod_007",
  "productName": "Brahmi",
  "inWishlist": true
}

order.status_update
{
  "orderId": "order_001",
  "previousStatus": "shipped",
  "newStatus": "out_for_delivery",
  "estimatedDelivery": "2026-01-25"
}
```

9. Success Metrics

9.1 Engagement Metrics

Metric	Target	Measurement
Section visits per week	> 2	Users visiting Shop section
Products viewed per visit	> 5	Product detail views
Time in section	> 3 min/ visit	Average session duration
Search usage	> 30%	Searches per visit

9.2 Conversion Metrics

Metric	Target	Measurement
Add to cart rate	> 15%	Products added vs. viewed
Cart conversion rate	> 40%	Checkouts vs. carts created
Average order value	> $50	Mean order total
Items per order	> 2	Average products per order

9.3 Personalization Metrics

Metric	Target	Measurement

Recommendation click rate	> 25%	Clicks on "For You" products
Protocol product purchase	> 50%	Protocol products bought
Match score accuracy	> 80%	User agreement with match scores
Personalized dosing usage	> 70%	Users who follow dosing

9.4 Quality Metrics

Metric	Target	Measurement
Return rate	< 5%	Products returned
Satisfaction rating	> 4.5/5	Post-purchase rating
Repeat purchase rate	> 30%	Users who reorder
Supplement adherence	> 60%	Users who take purchased supplements

9.5 Integration Metrics

Metric	Target	Measurement
Supplement section sync	> 90%	Purchases added to Supplements
Today task creation	> 80%	Tasks created from purchases
Protocol completion support	> 40%	Protocol products fully purchased
Health metric correlation	Positive	Health improvement with purchases

9.6 Technical Metrics

Metric	Target	Measurement
Shop load time	< 2s	Time to display shop home
Product load time	< 1s	Time to load product detail
Match score calculation	< 500ms	Time to calculate match
Search response time	< 1s	Time to return search results
Checkout completion	< 30s	Time to complete checkout
Error rate	< 1%	Failed operations

10. Implementation Considerations

10.1 Product Catalog Management

Product Data:

- Comprehensive Ayurvedic properties for each product
- Quality indicator data sourced/verified
- Regular price and inventory updates
- Product images optimized for mobile

Match Score Caching:

- Pre-calculate match scores when user profile changes
- Cache scores with TTL
- Recalculate on protocol changes
- Batch calculation for listing pages

10.2 E-Commerce Integration

Payment Processing:

- Secure payment gateway integration
- Multiple payment method support
- PCI compliance
- Saved payment methods

Inventory Management:

- Real-time stock levels
- Low stock alerts
- Backorder handling
- Multi-warehouse support

Shipping:

- Multiple shipping options
- Real-time rate calculation
- Tracking integration
- Delivery estimates

10.3 Post-Purchase Integration

Supplements Section:

- Auto-create supplement entries
- Include personalized dosing
- Link to product for reorder
- Track supply levels

Today Section:

- Create reminder tasks
- Schedule based on dosing
- Link to log functionality
- Expire when supply ends

Protocol Connection:

- Mark protocol products as purchased
- Track protocol completion

- Update protocol progress
- Suggest remaining products

10.4 Notification System

Order Notifications:

- Order confirmation
- Shipping updates
- Delivery confirmation
- Reorder reminders

Product Notifications:

- Price drops (wishlist items)
- Back in stock
- New products (matched to profile)
- Seasonal recommendations

Reminder Notifications:

- Low supply reminders
- Supplement taking reminders
- Protocol product suggestions

10.5 Offline Handling

When Offline:

- View cached products
- View cart
- View wishlist
- View order history
- Cannot complete purchase

When Back Online:

- Sync cart
- Update prices
- Process any queued actions
- Refresh recommendations

10.6 Accessibility Requirements

Screen Reader:

- Product information accessible
- Match scores announced
- Cart operations confirmed
- Checkout flow navigable

Motor Accessibility:

- Large tap targets for add to cart
- Swipe to add to wishlist
- Simple quantity controls
- Accessible checkout

Visual Accessibility:

- WCAG AA contrast
- Price clearly visible
- Match badges distinguishable
- Support Dynamic Type

10.7 Analytics Events

Event	Properties	Purpose
`shop_opened`	source, tab	Track entry
`product_viewed`	product_id, match_score	Product interest
`product_added_cart`	product_id, source	Conversion
`product_removed_cart`	product_id, reason	Drop-off
`cart_viewed`	item_count, total	Checkout intent
`checkout_started`	item_count, total	Conversion
`checkout_completed`	order_id, total	Conversion
`checkout_abandoned`	step, item_count	Drop-off
`search_performed`	query, results_count	Search usage
`filter_applied`	filter_type, value	Filter usage
`recommendation_clicked`	product_id, section	Personalization
`wishlist_added`	product_id	Interest
`match_score_viewed`	product_id, score	Personalization
`dosing_viewed`	product_id	Engagement

10.8 Testing Priorities

Unit Tests:

- Match score calculation
- Price calculations
- Filter logic
- Cart operations

Integration Tests:

- Product detail load with personalization
- Cart to checkout flow
- Post-purchase integrations
- Search and filter

E2E Tests:

- Browse to purchase flow
- Search to purchase flow
- Wishlist management
- Order tracking

Manual QA:

- Match score accuracy
- Personalized dosing relevance
- Product data completeness
- Quality indicator accuracy
- Ayurvedic properties correctness

Appendix A: Ayurvedic Properties Reference

Rasa (Taste) Effects on Doshas

Taste	Sanskrit	Vata	Pitta	Kapha	Elements	Qualities
Sweet	Madhura	↓	↓	↑	Earth+Water	Heavy, moist, cooling
Sour	Amla	↓	↑	↑	Earth+Fire	Light, moist, heating
Salty	Lavana	↓	↑	↑	Water+Fire	Heavy, moist, heating
Pungent	Katu	↑	↑	↓	Fire+Air	Light, dry, heating
Bitter	Tikta	↑	↓	↓	Air+Ether	Light, dry, cooling
Astringent	Kashaya	↑	↓	↓	Air+Earth	Light, dry, cooling

Common Ayurvedic Actions

Action	Sanskrit	Description	Common Products
Rejuvenation	Rasayana	Anti-aging, vitality	Ashwagandha, Chyawanprash, Shatavari
Digestive Stimulant	Deepana	Kindles Agni	Ginger, Trikatu, Hingvastak
Digestive	Pachana	Aids digestion	Triphala, CCF tea
Mind Support	Medhya	Cognitive function	Brahmi, Shankhpushpi, Gotu Kola
Sleep Promoting	Nidrajanana	Promotes sleep	Ashwagandha, Tagara, Jatamansi
Strength Building	Balya	Physical strength	Ashwagandha, Shatavari, Bala
Immunity Building	Ojas Vardhana	Builds Ojas	Chyawanprash, Amalaki, Ghee
Vata Calming	Vatahara	Reduces Vata	Ashwagandha, Sesame oil, Ghee
Pitta Cooling	Pittahara	Reduces Pitta	Shatavari, Amalaki, Coconut
Kapha Reducing	Kaphahara	Reduces Kapha	Trikatu, Ginger, Honey

Seasonal Product Recommendations

Season	Sanskrit	Qualities	Recommended Products
Late Winter	Shishira	Cold, dry	Warming oils, Chyawanprash, Trikatu
Spring	Vasanta	Warm, moist	Trikatu, Honey, Detox teas
Summer	Grishma	Hot, dry	Coconut oil, Shatavari, Cooling herbs
Monsoon	Varsha	Cool, wet	Digestive spices, Warming herbs
Autumn	Sharad	Cool, dry	Amalaki, Cooling herbs, Ghee
Early Winter	Hemanta	Cold, moist	Warming oils, Ashwagandha, Ghee

Appendix B: Quality Indicator Categories

Manufacturing Standards

Level	Criteria
Excellent	GMP certified, FDA registered, ISO certified
High	GMP certified, third-party audited
Good	GMP practices followed
Moderate	Basic manufacturing standards
Low	Unknown or uncertified

Testing Benchmarks

Level	Criteria
Excellent	Third-party tested, COA available, heavy metal tested
High	Third-party tested, COA available
Good	Internal testing, batch verification
Moderate	Basic testing
Low	Minimal or no testing disclosed

Product Certifications

Certification	Meaning
USDA Organic	Certified organic ingredients
Non-GMO Project	Verified non-GMO

GMP Certified	Good Manufacturing Practices
Vegan	No animal products
Gluten-Free	Certified gluten-free
Kosher	Kosher certified
Halal	Halal certified

End of Document